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Grid Generation

Use a pre-trained model to generate new text through weighted random sampling.

You will need

- your completed bigram model (i.e. your filled-out grid) from *Grid Training*
- d10 (or similar) for weighted sampling
- pen & paper for writing down the generated “output text”

Your goal

To generate new text from your bigram language model. **Stretch goal:** keep going, generating as much text as possible. Write a whole book!

Key idea

Language models generate text by predicting one word at a time based on learned patterns. Your trained model provides the “next word” options and their relative probabilities; dice rolls provide the randomness to choose one of those options (and this process can be repeated indefinitely).



Algorithm

1. **choose a starting word** — pick any word from the first column of your grid
2. **look at that word's row** to identify all possible next words and their counts
3. **roll dice weighted by the counts** (see the *Weighted Randomness* lesson)
4. **write down the chosen word** and use that as your next starting word
5. **repeat** from step 2 until you reach the desired length or a natural stopping point (e.g. a full stop .)

Example

Using the same bigram model from the example in *Grid Training*:

	see	spot	run	.	jump	,
see						
spot						
run						
.						
jump						
,						

- choose (for example) **see** as your starting word
- **see** (row) → **spot** (column); it's the only option, so write down **spot** as next word
- **spot** → **run** (25%), **jump** (25%) or **,** (50%); roll dice to choose
- let's say dice picks **run**; write it down
- **run** → **.** (67%) or **,** (33%); roll dice to choose
- let's say dice picks **.**; write it down
- **.** → **see** (33%), **run** (33%) or **jump** (33%); roll dice to choose
- let's say dice picks **see**; write it down
- **see** → **spot**; it's the only option, so write down **spot**... and so on

After the above steps, the generated text is “see spot run. see spot”